# Lecture Notes V

## Course: Analytical Mechanics

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### Topic: Rigid body dynamics

**Remark:** These notes are typed by hand using the mentioned references and the only parts done by AI are the figures (using TikZ).

**Abstract.** In this section we study the motion of rigid bodies using the Lagrangian and Hamiltonian formalisms. Let me give some remarks on notation:

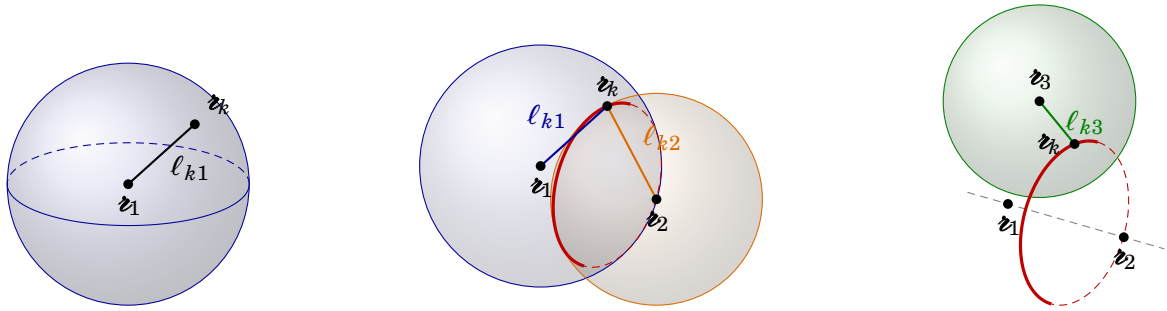
- The symbol $\mathbb{N}_N$ denotes the set $\{1,2,\dots,N\}$ for any $N \in \mathbb{N}$, and the notation $\sigma \in \mathbb{N}_N \times \mathbb{N}_N$ should be read as "$\sigma$ is a multi-index (2 values) with natural range".
- The symbol $\boldsymbol{r}_{A/B}$ denotes the position vector of position $A$ measured from a frame whose origin is the position $B$ with respect to some fixed inertial frame. All frames should be assumed inertial unless otherwise stated, for example when working with rotating frames.
- The letter $G$ should always be thought as to represent the centre of mass or a quantity related to the latter.
- For vectors, we will always use both the bold and the braket notation, i.e., $\mathbf{q} \equiv |q\rangle$.
- The notation $(A)^{\mathscr{I}}_{ij}$ represents the components of the transformation $A$ in the basis that defines the frame $\mathscr{I}$.

## Rigid Bodies

[1] A rigid body is a collection of particles constrained to be motionless relative to one another, put in another way, it is a multi-particle system which we assume has "rigid" interconnections between particles. More precisely (mathematically), a **rigid body** is a system of $N$ particles constrained by the equations:

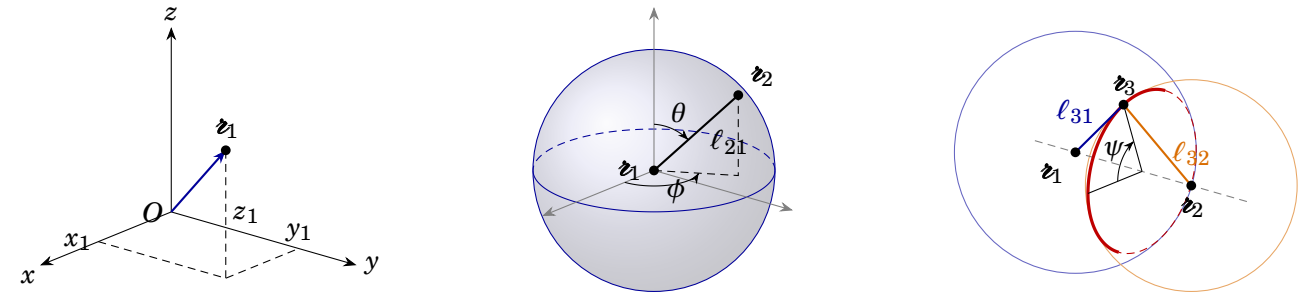
$$||\boldsymbol{r}_{i/j}|| = \ell_{ij}, \tag{1}$$

where $\ell_\sigma$ is a constant for each $\sigma \in \mathbb{N}_N \times \mathbb{N}_N$. [3] These constraints significantly reduce the degrees of freedom to six, fact which can be seen from the following argument.
First (see Figure 1), we only need to specify the position of three "reference" points (non-collinear), because once we have one point $\boldsymbol{r}_1$, any other point, say $\boldsymbol{r}_k$, in the body has a prescribed distance to $\boldsymbol{r}_1$, namely, $\ell_{k1}$ but this leaves the option for the point $\boldsymbol{r}_k$ to be in any point on a sphere of such radius, and if we also provide another point $\boldsymbol{r}_2$, then we also have a prescribed distance $\ell_{k2}$ so now the particle is in some circle and finally, if we provided a third point $\boldsymbol{r}_3$, the point in the latter circle is now fixed. We therefore conclude that any point in the rigid body can be specified by providing the position of three reference points.
Next (see Figure 2), to specify a reference point $\boldsymbol{r}_1$ in the rigid body we have to provide three coordinates, but then the other points $\boldsymbol{r}_2, \boldsymbol{r}_3$ in the body are constrained to move according to (1) with $j = 1$ fixed, and thus the position of $\boldsymbol{r}_2$ can be specified by providing two coordinates (angles in a sphere of radius $\ell_{21}$ around $\boldsymbol{r}_1$) and finally, the third point must lie in a circle and therefore only one more coordinate is needed to specify the position of $\boldsymbol{r}_3$, the conclusion is that six coordinates are needed to specify the complete configuration of a rigid body in space.




(a) $\boldsymbol{r}_1$ given: $\boldsymbol{r}_k$ lies on a sphere (b) $\boldsymbol{r}_1, \boldsymbol{r}_2$ given: $\boldsymbol{r}_k$ lies on a circle (c) $\boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3$ given: $\boldsymbol{r}_k$ is fixed

Figure 1: Three reference points determine every point of the rigid body.




(a) $\boldsymbol{r}_1$: three coordinates $(x_1, y_1, z_1)$ (b) $\boldsymbol{r}_2$: two angles $(\theta, \phi)$ (c) $\boldsymbol{r}_3$: one angle $\psi$

Figure 2: Six coordinates fix the three reference points.

## The Instantaneous Angular Velocity and the Inertia Tensor

Let us now consider the case of a rigid body with a fixed point, this means that one of the particles $\boldsymbol{r}_f$ is fixed with respect to some inertial frame $\mathscr{R} = \{O, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$. By an appropriate transformation of $\mathscr{R}$, we can get a new inertial frame with origin at $\boldsymbol{r}_f$, namely, $\mathscr{I} = \{\boldsymbol{r}_f, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is now considered as the "fixed" frame, and we will also define a new (orthogonal) set of axes $\mathscr{B} = \{O', \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ fixed in the body and sharing the origin with $\mathscr{I}$, i.e., $O' = \boldsymbol{r}_f$, then we can relate these two coordinates by a transformation $\mathsf{R} \in SO(3), \mathsf{R} = \mathsf{R}(t)$ in such way that

$$|b_i\rangle = \mathsf{R}\,|e_i\rangle\,,$$

and since we require that the new set of axes is also orthonormal (provided that $\mathscr{I}$ is) we have the requirement

$$\langle b_i | b_j \rangle = \langle e_i | \mathsf{R}^\dagger \mathsf{R} | e_j \rangle = \langle e_i | e_j \rangle\,,$$

so that $\mathsf{R}^\dagger \mathsf{R} = \mathsf{I}$, i.e., $\mathsf{R}$ is unitary. In particular, $\mathsf{R}$ is orthogonal (with $\det \mathsf{R} = 1$ we require the new body-attached frame to stay right-handed) because it is real so that there is some antisymmetric matrix $\Omega$ for which ${}^{\mathscr{I}}\mathrm{d}\mathsf{R}/\mathrm{d}t = \Omega\mathsf{R}$ where we write $(\Omega)^{\mathscr{I}}_{ij} = -\epsilon_{ijk}\omega_k$ for some ${}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{B}}$, called the **instantaneous angular velocity**. Using the transport equation we have

$${}^{\mathscr{I}}\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{\alpha/f} = {}^{\mathscr{I}}\mathbf{v}_{\alpha/f} = {}^{\mathscr{B}}\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{\alpha/f} + {}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{B}} \times \boldsymbol{r}_{\alpha/f} = {}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{B}} \times \boldsymbol{r}_{\alpha/f}\,,$$

where ${}^{\mathscr{B}}\mathrm{d}\boldsymbol{r}_{\alpha/f}/\mathrm{d}t = 0$ since the $\mathscr{B}$ frame is fixed to the body and thus the particles are static with respect to $\mathscr{B}$. Using this, we see that the kinetic energy of the system (measured from $\mathscr{I}$) is given by

$$T = \frac{1}{2}\sum_{\alpha=1}^{N} m_\alpha\, {}^{\mathscr{I}}\mathbf{v}^2_{\alpha/f} = \frac{1}{2}\sum_{\alpha=1}^{N} m_\alpha \Big({}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{B}} \times \boldsymbol{r}_{\alpha/f}\Big) \cdot \Big({}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{B}} \times \boldsymbol{r}_{\alpha/f}\Big),$$

which on writing $\boldsymbol{r}_{\alpha/f} = \xi^{(\alpha)}_i |b_i\rangle$ and ${}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{B}} = \omega_i |b_i\rangle$ and recalling that $\langle b_i | b_j \rangle = \delta_{ij}$ and $\mathbf{b}_i \times \mathbf{b}_j = \varepsilon_{ijk}\mathbf{b}_k$, i.e., $\mathscr{B}$ is a right-handed frame, we have

$$T = \frac{1}{2}\sum_{\alpha=1}^{N} m_\alpha \varepsilon_{ijk}\omega_j \xi^{(\alpha)}_k \varepsilon_{ilm}\omega_l \xi^{(\alpha)}_m = \frac{1}{2}\sum_{\alpha=1}^{N} m_\alpha \big(\delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}\big)\omega_j \xi^{(\alpha)}_k \omega_l \xi^{(\alpha)}_m = \frac{1}{2}\sum_{\alpha=1}^{N} m_\alpha \Big(\omega_j \xi^{(\alpha)}_k \omega_j \xi^{(\alpha)}_k - \omega_j \xi^{(\alpha)}_k \omega_k \xi^{(\alpha)}_j\Big),$$

and, as it is customary [2], we factorise the terms $\omega_j\omega_k$:

$$T=\frac{1}{2}\sum_{\alpha=1}^{N}m_\alpha\left(\omega_j\boldsymbol{\varrho}_{\alpha/f}^2\omega_k\delta_{jk}-\omega_j\xi_k^{(\alpha)}\omega_k\xi_j^{(\alpha)}\right)=\frac{1}{2}\omega_j\omega_k\sum_{\alpha=1}^{N}m_\alpha\left(\boldsymbol{\varrho}_{\alpha/f}^2\delta_{jk}-\xi_j^{(\alpha)}\xi_k^{(\alpha)}\right),$$

where the nine real numbers $I_{jk}=\sum_{\alpha=1}^{N}m_\alpha\left(\boldsymbol{\varrho}_{\alpha/f}^2\delta_{jk}-\xi_j^{(\alpha)}\xi_k^{(\alpha)}\right)$ form the components of what we call the **inertia tensor** of the rigid body, and using this definition we get

$$T=\frac{1}{2}\omega_j\omega_k I_{jk}^{(f)},$$

[1] the numbers $I_{ii}$ are called **moments of inertia** and the elements $I_{ij},i\neq j$ **products of inertia**. Notice that the numbers $I_{jk}$ are constant since we are expressing the position of the particles measured from the body-attached frame, where the positions are constant. The previous result is sometimes written

$$T=\frac{1}{2}\left({}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{B}}\right)^{\dagger}\mathbf{I}_{\mathscr{B}}^{(f)}\,{}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{B}}\equiv\frac{1}{2}\boldsymbol{\omega}^{\dagger}\mathbf{I}^{(f)}\boldsymbol{\omega},$$

with $\mathbf{I}=(I_{jk})$. For the case of a continuous distribution $\rho$ the components of the inertia tensor $\mathbf{I}_{\mathscr{F}}=(I_{jk}^{\mathscr{F}})$ in a (not necessarily inertial) reference frame $\mathscr{F}$ are immediately generalised to

$$(I_{jk}^{\mathscr{F}})=\int_V \mathrm{d}^3\mu\,\rho(\boldsymbol{\varrho})\left(\delta_{jk}\boldsymbol{\varrho}^2-r_jr_k\right),$$

where $\mathscr{F}=(P,\boldsymbol{\eta}_1,\boldsymbol{\eta}_2,\boldsymbol{\eta}_3)$ and $\boldsymbol{\varrho}=r_i\left|\eta_i\right\rangle$. Going back to the inertial frame $\mathscr{R}=\{O,\mathbf{e}_1,\mathbf{e}_2,\mathbf{e}_3\}$ notice that the position of any particle in the body can be written as

$$\boldsymbol{\varrho}_{\alpha/O}=\boldsymbol{\varrho}_{\alpha/f}+\boldsymbol{\varrho}_{f/O}\tag{2}$$

but since the point $\boldsymbol{\varrho}_{f/O}$ is fixed when measured from $\mathscr{R}$, we have ${}^{\mathscr{R}}\mathrm{d}\boldsymbol{\varrho}_{f/O}/\mathrm{d}t=0$ and hence

$$\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{\alpha/O}=\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{\alpha/f}$$

and since $\mathscr{R}$ and $\mathscr{I}$ have the same set of axes (they only differ by their fixed origins), then

$$\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{\alpha/O}=\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{\alpha/f}=\frac{{}^{\mathscr{I}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{\alpha/f},$$

which shows that all the results relating derivatives are the same when considering measurements from $\mathscr{R}$.

## Decomposition of the Kinetic Energy (Primarily from [1])

In the most general case we would have that the "fixed" point $\boldsymbol{\varrho}_f$ in the body is now not fixed with respect to $\mathscr{R}$, let us choose this new moving reference point as the centre of mass $\boldsymbol{\varrho}_{G/O}$ so that the derivatives are related by

$$\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{\alpha/O}=\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{\alpha/G}+\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{G/O}\equiv\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{\alpha/G}+{}^{\mathscr{R}}\mathbf{v}_{G/O}$$

according to (2), where ${}^{\mathscr{R}}\mathbf{v}_{G/O}$ is the translational velocity of the centre of mass and the other term is just the velocity of the particle measured from the centre of mass, meaning that when taking the body-attached frame, we get the relation (writing all the derivatives explicitly)

$$\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{\alpha/O}=\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{\alpha/G}+\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{G/O}=\frac{{}^{\mathscr{G}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{\alpha/G}+{}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{G}}\times\boldsymbol{\varrho}_{\alpha/G}+\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{G/O}$$

where $\mathscr{G}=\{\boldsymbol{\varrho}_{G/O},\mathbf{b}_1,\mathbf{b}_2,\mathbf{b}_3\}$ is the body-attached frame centred at the centre of mass and therefore (as we mentioned before) ${}^{\mathscr{G}}\mathrm{d}\boldsymbol{\varrho}_{\alpha/G}/\mathrm{d}t=0$ and the velocity of the particles is thus given by

$$\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{\alpha/O}={}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{G}}\times\boldsymbol{\varrho}_{\alpha/G}+\frac{{}^{\mathscr{R}}\mathrm{d}}{\mathrm{d}t}\boldsymbol{\varrho}_{G/O}$$

and letting $\mathscr{I}$ be the (non-rotating but also non-inertial) frame with the same axes as $\mathscr{R}$ but with the origin shifted to the centre of mass, we get that ${}^{\mathscr{R}}\mathrm{d}/\mathrm{d}t = {}^{\mathscr{I}}\mathrm{d}/\mathrm{d}t$. We now compute the kinetic energy:

$$T = \frac{1}{2}\sum_{\alpha=1}^{N} m_\alpha \left( {}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{G}} \times \boldsymbol{r}_{\alpha/G} + {}^{\mathscr{I}}\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{G/O} \right)^2 = \frac{1}{2}\sum_{\alpha=1}^{N} m_\alpha \left( {}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{G}} \times \boldsymbol{r}_{\alpha/G} \right)^2 + \frac{1}{2}\sum_{\alpha=1}^{N} m_\alpha \left( {}^{\mathscr{I}}\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{G/O} \right)^2 + \underbrace{\sum_{\alpha=1}^{N} m_\alpha \left( {}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{G}} \times \boldsymbol{r}_{\alpha/G} \right) \cdot {}^{\mathscr{I}}\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{G/O}}_{X},$$

where we now claim that

$$\sum_{\alpha=1}^{N} m_\alpha \boldsymbol{r}_{\alpha/G} = 0$$

which can be seen from the fact that

$$\boldsymbol{r}_{\alpha/O} = \boldsymbol{r}_{G/O} + \boldsymbol{r}_{\alpha/G}$$

so that

$$\sum_{\alpha=1}^{N} m_\alpha \boldsymbol{r}_{\alpha/O} = \sum_{\alpha=1}^{N} m_\alpha \boldsymbol{r}_{G/O} + \sum_{\alpha=1}^{N} m_\alpha \boldsymbol{r}_{\alpha/G}$$

but by definition of centre of mass one has

$$\boldsymbol{r}_{G/O} = \frac{\sum_{\alpha=1}^{N} m_\alpha \boldsymbol{r}_{\alpha/O}}{\sum_{\alpha=1}^{N} m_\alpha}$$

so $\sum_{\alpha=1}^{N} m_\alpha \boldsymbol{r}_{\alpha/G} = 0$. Then the cross term $X$ becomes

$$X = {}^{\mathscr{I}}\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{G/O} \cdot \left( {}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{G}} \times \sum_{\alpha=1}^{N} m_\alpha \boldsymbol{r}_{\alpha/G} \right) = {}^{\mathscr{I}}\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{G/O} \cdot \left( {}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{G}} \times 0 \right) = 0$$

and following a similar procedure as in the previous section, the kinetic energy is simply

$$T = \frac{1}{2}\sum_{\alpha=1}^{N} m_\alpha \left( {}^{\mathscr{I}}\boldsymbol{\omega}^{\mathscr{G}} \times \boldsymbol{r}_{\alpha/G} \right)^2 + \frac{1}{2}\sum_{\alpha=1}^{N} m_\alpha \left( {}^{\mathscr{R}}\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{r}_{G/O} \right)^2 = \frac{1}{2}\omega_i \omega_j I_{ij}^{(G)} + \frac{1}{2}M\,{}^{\mathscr{I}}\mathbf{V}_{G/O}^2,$$

where the first term is the kinetic energy due to rotations about the centre of mass and the second term is the usual kinetic energy of the object considered as one particle of mass $\sum_\alpha m_\alpha$ located at the centre of mass.

## Worked out example [2]

*Example (Inertia tensor of a homogeneous cylinder).* Let us consider a homogeneous right circular cylinder. The height of the cylinder will be denoted by $h$, its radius by $a$, and its mass by $M$. Then its density, assumed constant, is $\rho_0 = M/(\pi a^2 h)$. We will compute the inertia tensor about the centre of mass (which is left as an exercise to prove that is the centre of the cylinder) and we will make the $\mathbf{b}_3$ axis to coincide with the axis of the cylinder. We can thus compute the inertia tensor $\mathbf{I}_{\mathscr{B}} = (I_{ij})$ (where the components are constant) using cylindrical coordinates, to do so we write $\mathscr{B} = (G, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ $\boldsymbol{r} = \eta_i |b_i\rangle$ and $\eta_1 = r\cos\phi, \eta_2 = r\sin\phi, \eta_3 = z$

$$I_{11}^{\mathscr{B}} = \int \mathrm{d}^3\mu\, \rho_0(\eta_1^2 + \eta_2^2 + \eta_3^2 - \eta_1^2) = \int \mathrm{d}^3\mu\, \rho_0(\eta_2^2 + \eta_3^2) = \int_0^a \int_0^{2\pi} \int_{-h/2}^{h/2} \mathrm{d}z\mathrm{d}\phi\mathrm{d}r\; \rho_0 r(r^2\sin^2\phi + z^2)$$

$$= \int_0^a \int_0^{2\pi} \int_{-h/2}^{h/2} \mathrm{d}z\mathrm{d}\phi\mathrm{d}r\; \rho_0(r^3\sin^2\phi + rz^2) = \rho_o \int_0^a \int_{-h/2}^{h/2} \mathrm{d}z\mathrm{d}r\left(r^3\pi + 2\pi r z^2\right) = \rho_0\pi \int_{-h/2}^{h/2} \mathrm{d}z \left( \frac{a^4}{4} + z^2 a^2 \right)$$

$$= \rho_0\pi\left( \frac{a^4 h}{4} + \frac{h^3 a^2}{12} \right) = \rho_0 \pi a^2 h \left( \frac{a^2}{4} + \frac{h^2}{12} \right) = M\left( \frac{a^2}{4} + \frac{h^2}{12} \right).$$

The other terms are found in a similar fashion, and using the parity of the integrands in the non-diagonal terms we get

$$\mathbf{I}_{\mathscr{B}} = \frac{M}{12}\begin{pmatrix} 3a^2 + h^2 & 0 & 0 \\ 0 & 3a^2 + h^2 & 0 \\ 0 & 0 & 6a^2 \end{pmatrix}.$$

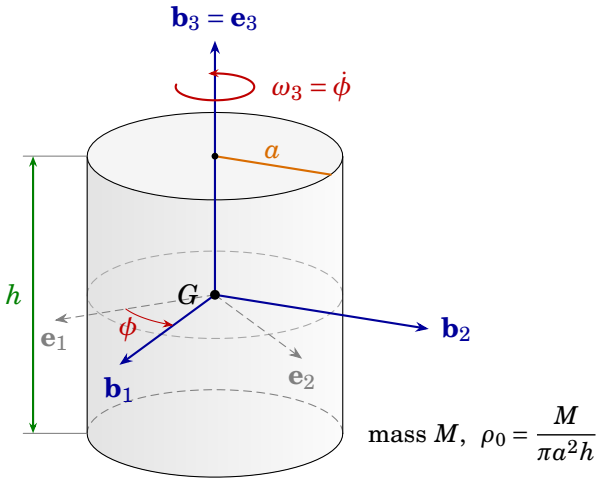



Figure 3: Homogeneous right circular cylinder of Example 1.

In this example one can define the rotation matrix around the $z$-axis by

$$\mathsf{R} = \begin{pmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

so that

$$\dot{\mathsf{R}}\mathsf{R}^{\dagger} = -\dot{\phi}\begin{pmatrix} \sin\phi & \cos\phi & 0 \\ -\cos\phi & \sin\phi & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -\dot{\phi} & 0 \\ \dot{\phi} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \Omega$$

and with the aid of definition $(\Omega)_{ij} = -\varepsilon_{ijk}\omega_k$ we get that $\omega_1 = \omega_2 = 0$ and $\omega_3 = \dot{\phi}$ and thus the kinetic energy due to rotations is

$$T = \frac{1}{2} I_{33}\omega_3\omega_3 = \frac{1}{4} M a^2 \dot{\phi}^2.$$

## Principal Axis Frame

It is important to note that the fact that $I$ is a diagonal matrix, greatly simplifies the computation of $T$, this is not always the case, once we choose a body-attached frame $\mathscr{B} = (G, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ we may or may not get a diagonal matrix, but if we don't, since $\mathbf{I}_{\mathscr{B}}$ is symmetric, we can always diagonalise $\mathbf{I}_{\mathscr{B}}$, i.e., find a new basis $\mathscr{E} = (O, \mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3)$ where $\mathbf{I}_{\mathscr{E}} = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$ and $\mathbf{I}_{\mathscr{B}}\mathbf{n}_i = \lambda_i \mathbf{n}_i$ (no sum), i.e., $\lambda_i$ is the eigenvalue of the eigenvector $\mathbf{n}_i$, which we know from linear algebra form an orthogonal basis. In such a new frame, the inertia tensor is always diagonal, such a special frame, "the best body-attached frame" is called the **principal axis frame** and the eigenvalues $\lambda_i = I^{\mathscr{E}}_{ii}$ are called the **principal moments of inertia**. The indices of the eigenvectors of the new basis are chosen in such way that

$$\mathbf{n}_i \times \mathbf{n}_j = \varepsilon_{ijk}\mathbf{n}_k.$$

[4] There is a relation between the centre of mass and the inertia tensor that is worth noting. There are three related quantities that describe the mass distribution

| Question | Answer | Expression |
|---|---|---|
| *How much?* | Zeroth moment (mass) | $\int_V d^3\mu\,\rho(\boldsymbol{r})$ |
| *Where is it?* | First moment (centre of mass) | $\boldsymbol{r}_{G/O} = \frac{1}{M}\int_V d^3\mu\,\rho(\boldsymbol{r})\,\boldsymbol{r}$ |
| *How spread is it?* | Second moment (inertia tensor) | $I_{jk}^{(G)} = \int_V \rho(\boldsymbol{r})\left(\delta_{jk}\,\boldsymbol{r}^2 - r_j r_k\right)$ |

Table 1: The three moments of the mass distribution of a rigid body.

## References


[1] N. J. Kasdin and D. A. Paley, *Engineering Dynamics: A Comprehensive Introduction*, Princeton University Press, Princeton (2011).

[2] G. F. Torres del Castillo, *An Introduction to Hamiltonian Mechanics*, Birkhäuser, Cham (2018).

[3] H. Goldstein, C. P. Poole and J. L. Safko, *Classical Mechanics*, 3rd ed., Addison-Wesley, San Francisco (2002).

[4] S. D. Ross, *Ross Dynamics Lab* [YouTube channel], Virginia Tech, Blacksburg. See in particular the lecture series *Analytical Dynamics: Lagrangian and 3D Rigid Body Dynamics* (2020). https://www.youtube.com/c/RossDynamicsLab (accessed August 2026).